

Pregnancy Tracker System - Full Stack Development Prompt

منصة وداد الصحية - نظام متتبع الحمل

📋 Overview - نظرة عامة

هذا البرمبت لبناء نظام متتبع الحمل الكامل في منصة وداد الصحية، يشمل:

- تتبع مراحل الحمل أسبوعيًا (40 أسبوع)
- معلومات مخصصة لكل أسبوع
- متابعة الوزن والعلامات الحيوية
- سجل الأعراض والفحوصات
- نصائح غذائية وطبية
- (لاحقًا) AI التكامل مع

🎯 Project Context - السياق

الأجزاء المكتملة:

- ✅ نظام المصادقة (Authentication)
- ✅ نظام الملف الشخصي (Patient Profile)
- ✅ Health Trackers (Mood, Weight, Period, Fertility)

الجزء المطلوب بناؤه:

🔑 نظام متتبع الحمل (Pregnancy Tracker)

🗄 Database Schema (Already Exists - للمراجعة)

1. pregnancies

```
sql
```

- id
- patient_id (FK → patients.id)
- conception_date (date, nullable)
- last_menstrual_period (date) - LMP
- due_date (date) - calculated or manual
- current_week (integer, nullable) - 1 to 42
- is_active (boolean, default: true)
- pregnancy_status (enum: 'ongoing', 'completed', 'miscarriage', 'terminated')
- delivery_date (date, nullable)
- delivery_type (enum: 'normal', 'cesarean', nullable)
- notes (text, nullable)
- timestamps

2. pregnancy_entries

sql

- id
- pregnancy_id (FK → pregnancies.id)
- week_number (integer) - 1 to 42
- weight (decimal 5,2, nullable) - kg
- blood_pressure_systolic (integer, nullable) - mmHg
- blood_pressure_diastolic (integer, nullable) - mmHg
- symptoms (json, nullable) - array of symptoms
- notes (text, nullable)
- entry_date (date)
- timestamps
- index(pregnancy_id, week_number)

3. patient_medical_files (للاستخدام)

sql

- id
- patient_id (FK → patients.id)
- pregnancy_id (FK → pregnancies.id, nullable)
- file_name
- file_path
- file_type (pdf, jpg, png, dicom)
- file_size (bytes)
- category (enum: 'lab_result', 'ultrasound', 'x_ray', 'prescription', 'medical_report', 'other')
- description (text, nullable)
- file_date (date, nullable)
- uploaded_by (FK → doctors.id, nullable)
- timestamps
- index(patient_id, category)

Backend API Endpoints

Pregnancy Management

POST /api/v1/patient/pregnancy/start

بدء حمل جديد

Headers: Authorization: Bearer {token}

Middlewares: auth:patient, PatientStatus, PatientEmailVerify

Request Body:

```
{
  "last_menstrual_period": "2025-10-01", // required
  "conception_date": "2025-10-15", // optional
  "due_date": "2026-07-08", // optional, calculated if not provided
  "notes": "حمل أول" // optional
}
```

Validation:

- last_menstrual_period: required|date|before_or_equal:today
- conception_date: nullable|date|after:last_menstrual_period
- due_date: nullable|date|after:last_menstrual_period
- notes: nullable|string|max:1000

Business Logic:

- If due_date not provided: calculate as LMP + 280 days (40 weeks)
- If conception_date not provided: estimate as LMP + 14 days
- Calculate current_week based on LMP and today
- Set previous pregnancy (if exists) as is_active = false
- Update patient's life_stage_id to 'pregnancy' (id = 3 في life_stages)

Response (201):

```
{
  "status": true,
  "message": "Pregnancy started successfully",
  "data": {
    "id": 1,
    "patient_id": 1,
    "last_menstrual_period": "2025-10-01",
    "conception_date": "2025-10-15",
    "due_date": "2026-07-08",
    "current_week": 17,
    "days_pregnant": 119,
    "weeks_remaining": 23,
    "days_remaining": 161,
    "trimester": 2,
    "is_active": true,
    "pregnancy_status": "ongoing",
    "notes": "حمل أول",
    "created_at": "2026-01-30 10:00:00"
  }
}
```

```
}  
}
```

GET /api/v1/patient/pregnancy/current

جلب الحمل النشط الحالي

Response (200):

```
{  
  "status": true,  
  "message": "Current pregnancy retrieved successfully",  
  "data": {  
    "id": 1,  
    "last_menstrual_period": "2025-10-01",  
    "due_date": "2026-07-08",  
    "current_week": 17,  
    "current_day": 3,  
    "days_pregnant": 119,  
    "weeks_remaining": 23,  
    "days_remaining": 161,  
    "trimester": 2,  
    "trimester_progress": 60, // percentage  
    "pregnancy_status": "ongoing",  
    "is_high_risk": false, // from AI predictions  
    "baby_development": {  
      "size_comparison": "الرمانة", // fruit/vegetable comparison  
      "length_cm": 13,  
      "weight_grams": 150  
    },  
    "next_milestone": {  
      "week": 20,  
      "description": "فحص الموجات فوق الصوتية التفصيلي"  
    },  
    "stats": {  
      "total_entries": 15,  
      "last_entry_date": "2026-01-28",  
      "last_weight": 68.5,  
      "weight_gained": 5.5  
    }  
  }  
}
```

Error (404): "No active pregnancy found"

PUT /api/v1/patient/pregnancy/{id}/complete

إنهاء الحمل (الولادة)

Request Body:

```
{
  "delivery_date": "2026-07-08", // required
  "delivery_type": "normal", // required, enum: normal, cesarean
  "notes": "ولادة طبيعية بدون مضاعفات" // optional
}
```

Validation:

- delivery_date: required|date|after:conception_date|before_or_equal:today
- delivery_type: required|in:normal,cesarean
- notes: nullable|string|max:2000

Response (200):

```
{
  "status": true,
  "message": "Pregnancy completed successfully. Congratulations! 🎉",
  "data": {
    "id": 1,
    "pregnancy_status": "completed",
    "delivery_date": "2026-07-08",
    "delivery_type": "normal",
    "pregnancy_duration_weeks": 40,
    "is_active": false,
    "notes": "..."
  }
}
```

GET /api/v1/patient/pregnancy/history

سجل الحمول السابقة

Response (200):

```
{
  "status": true,
  "data": {
    "pregnancies": [
      {
        "id": 1,
        "lmp": "2025-10-01",
        "due_date": "2026-07-08",
        "delivery_date": "2026-07-08",
        "delivery_type": "normal",
        "pregnancy_status": "completed",
        "duration_weeks": 40,
        "is_active": false
      }
    ],
    "stats": {
      "total_pregnancies": 2,
      "completed": 1,
      "ongoing": 1,
      "average_duration": 39.5
    }
  }
}
```

Pregnancy Entries (Weekly/Daily Tracking)

POST /api/v1/patient/pregnancy/entry

إضافة تسجيل أسبوعي

Request Body:

```
{
  "week_number": 17, // optional, defaults to current_week
  "weight": 68.5, // optional
  "blood_pressure_systolic": 120, // optional
  "blood_pressure_diastolic": 80, // optional
  "symptoms": [ // optional
    "morning_sickness",
    "back_pain",
    "fatigue",
    "heartburn"
  ],
  "notes": "اشعرت بحركة الجنين اليوم", // optional
  "entry_date": "2026-01-30" // optional, defaults to today
}
```

Validation:

- week_number: nullable|integer|min:1|max:42
- weight: nullable|numeric|min:30|max:200
- blood_pressure_systolic: nullable|integer|min:60|max:200
- blood_pressure_diastolic: nullable|integer|min:40|max:140
- symptoms: nullable|array
- notes: nullable|string|max:2000
- entry_date: nullable|date|before_or_equal:today

Response (201):

```
{
  "status": true,
  "message": "Pregnancy entry added successfully",
  "data": {
    "id": 1,
    "pregnancy_id": 1,
    "week_number": 17,
    "weight": 68.5,
    "weight_change": "+0.5", // from last entry
    "blood_pressure": "120/80",
    "bp_category": "normal", // normal, elevated, high
    "symptoms": ["morning_sickness", "back_pain"],
    "notes": "اشعرت بحركة الجنين اليوم",
    "entry_date": "2026-01-30",
    "alerts": [] // e.g., "ارتفاع ضغط الدم - استشير طبيبك"
```



```
}  
}
```

GET /api/v1/patient/pregnancy/entries

جلب التسجيلات

Query Parameters:

- week_number: optional, specific week
- start_week: optional
- end_week: optional

Response (200):

```
{  
  "status": true,  
  "data": {  
    "entries": [  
      {  
        "id": 1,  
        "week_number": 17,  
        "weight": 68.5,  
        "blood_pressure": "120/80",  
        "symptoms": [...],  
        "notes": "...",  
        "entry_date": "2026-01-30",  
        "has_alerts": false  
      }  
    ],  
    "summary": {  
      "total_entries": 15,  
      "weeks_tracked": [12, 13, 14, 15, 16, 17],  
      "average_weight": 67.5,  
      "total_weight_gain": 5.5,  
      "most_common_symptoms": ["fatigue", "back_pain"]  
    }  
  }  
}
```

GET /api/v1/patient/pregnancy/week/{number}

معلومات أسبوع محدد

Response (200):

```
{
  "status": true,
  "data": {
    "week_number": 17,
    "trimester": 2,
    "user_entries": [ // التسجيلات الخاصة بالمستخدم لهذا الأسبوع
      {
        "id": 1,
        "weight": 68.5,
        "blood_pressure": "120/80",
        "symptoms": [...],
        "entry_date": "2026-01-30"
      }
    ],
    "week_info": {
      "title": "الأسبوع 17 من الحمل",
      "baby_development": {
        "size_comparison": "الرمانة",
        "length_cm": 13,
        "weight_grams": 150,
        "developments": [
          "يمكن للجنين سماع الأصوات",
          "تتشكل طبقة الدهون تحت الجلد",
          "يبدأ الجنين في الحركة بشكل أكثر"
        ]
      },
      "mother_changes": [
        "قد تشعرين بحركة الجنين (الركلات الأولى)",
        "زيادة في الوزن بمعدل 0.5 كجم أسبوعيًا",
        "ظهور خط داكن في منتصف البطن (الخط الأسود)"
      ],
      "symptoms_to_expect": [
        "آلام الظهر",
        "حرقة المعدة",
        "تورم القدمين"
      ],
      "medical_tips": [
        "تناولي الأطعمة الغنية بالحديد",
        "احرصي على شرب الماء بكثرة",
        "تجنبي الوقوف لفترات طويلة"
      ],
      "nutrition_tips": [
```

```
    "احرصي على تناول الخضروات الورقية",
    "تناولي الحليب ومنتجات الألبان",
    "الأطعمة الغنية بأوميغا 3"
  ],
  "exercise_tips": [
    "المشي لمدة 30 دقيقة يوميًا",
    "تمارين قاع الحوض",
    "اليوجا للحوامل (تحت إشراف)"
  ],
  "warning_signs": [
    "نزيف مهبل",
    "آلام شديدة في البطن",
    "ارتفاع حاد في ضغط الدم"
  ],
  "checklist": [
    {
      "item": "فحص السكر في الدم",
      "recommended_week": "24-28",
      "is_due": false
    }
  ]
}
```

PUT /api/v1/patient/pregnancy/entry/{id}

تحديث تسجيل

DELETE /api/v1/patient/pregnancy/entry/{id}

حذف تسجيل

Pregnancy Statistics & Charts

GET /api/v1/patient/pregnancy/stats

إحصائيات الحمل

Response (200):

```
{
  "status": true,
  "data": {
    "weight_stats": {
      "starting_weight": 63.0,
      "current_weight": 68.5,
      "total_gain": 5.5,
      "average_weekly_gain": 0.35,
      "recommended_total_gain": "11-16", // based on pre-pregnancy BMI
      "is_on_track": true
    },
    "blood_pressure_stats": {
      "average_systolic": 118,
      "average_diastolic": 78,
      "highest_reading": "125/85",
      "lowest_reading": "110/70",
      "concern_level": "normal" // normal, monitor, concern
    },
    "symptoms_stats": {
      "total_unique_symptoms": 12,
      "most_frequent": [
        {"symptom": "fatigue", "count": 15},
        {"symptom": "back_pain", "count": 12},
        {"symptom": "heartburn", "count": 10}
      ]
    },
    "milestones_completed": [
      {
        "week": 12,
        "milestone": "نهاية الثلث الأول",
        "date": "2025-12-24"
      }
    ],
    "upcoming_milestones": [
      {
        "week": 20,
        "milestone": "فحص الموجات فوق الصوتية التفصيلي",
        "in_weeks": 3
      }
    ]
  }
}
```

```
}  
}
```

GET /api/v1/patient/pregnancy/weight-chart

بيانات الرسم البياني للوزن

Response (200):

```
{  
  "status": true,  
  "data": {  
    "chart_data": [  
      {  
        "week": 12,  
        "weight": 63.5,  
        "date": "2025-12-24"  
      },  
      {  
        "week": 13,  
        "weight": 64.0,  
        "date": "2025-12-31"  
      }  
    ],  
    "recommended_range": {  
      "min": [63.0, 63.5, 64.0, ...], // per week  
      "max": [64.0, 65.0, 66.0, ...]  
    },  
    "current_status": "within_range" // below, within_range, above  
  }  
}
```

Medical Files for Pregnancy

POST /api/v1/patient/pregnancy/files/upload

رفع ملف طبي مرتبط بالحمل

Headers:

- Content-Type: multipart/form-data
- Authorization: Bearer {token}

Request Body (FormData):

- file: required, file, mimes:pdf,jpg,jpeg,png, max:10240 (10MB)
- category: required, enum(lab_result, ultrasound, x_ray, prescription, medical_report, other)
- description: optional, string
- file_date: optional, date

Response (201):

```
{
  "status": true,
  "message": "File uploaded successfully",
  "data": {
    "id": 1,
    "file_name": "ultrasound_week_20.jpg",
    "file_url": "https://domain.com/storage/medical-files/...",
    "file_type": "jpg",
    "file_size": 2048576, // bytes
    "file_size_formatted": "2.0 MB",
    "category": "ultrasound",
    "description": "صورة سونار الأسبوع 20",
    "file_date": "2026-01-30",
    "pregnancy_id": 1,
    "uploaded_at": "2026-01-30 15:30:00"
  }
}
```

GET /api/v1/patient/pregnancy/files

جلب الملفات الطبية للحمل الحالي

Query Parameters:

- category: optional, filter by category
- pregnancy_id: optional, default: current active pregnancy

Response (200):

```
{
  "status": true,
  "data": {
    "files": [
      {
        "id": 1,
        "file_name": "ultrasound_week_20.jpg",
        "file_url": "...",
        "category": "ultrasound",
        "description": "...",
        "file_date": "2026-01-30",
        "file_size_formatted": "2.0 MB",
        "uploaded_at": "2026-01-30 15:30:00"
      }
    ],
    "grouped_by_category": {
      "ultrasound": 5,
      "lab_result": 3,
      "prescription": 2,
      "medical_report": 1
    }
  }
}
```

DELETE /api/v1/patient/pregnancy/files/{id}

حذف ملف طبي

Pregnancy Content & Information

GET /api/v1/pregnancy/weeks-info

(Public or Authenticated) معلومات جميع الأسابيع

Response (200):

```
{
  "status": true,
  "data": {
    "weeks": [
      {
        "week_number": 1,
        "trimester": 1,
        "baby_size": "حجم بذرة",
        "summary": "...",
        "image_url": "..."
      },
      // ... 40 weeks
    ]
  }
}
```

Frontend Requirements

Technology Stack:

- React 18+
- Recharts (للرسوم البيانية)
- Swiper (للتمرير بين الأسابيع)
- React Dropzone (لرفع الملفات)
- Date-fns (للتعامل مع التواريخ)
- Framer Motion (للتحريكات)

Pages Structure

```
src/
├── pages/
│   ├── patient/
│   │   ├── pregnancy/
│   │   │   ├── PregnancyDashboard.jsx (الصفحة الرئيسية)
│   │   │   ├── StartPregnancy.jsx (بدء حمل جديد)
│   │   │   ├── WeekByWeek.jsx (التصفح حسب الأسبوع)
│   │   │   └── WeekDetails.jsx (تفاصيل أسبوع محدد)
```




Main Pages Specifications

PregnancyDashboard.jsx (الصفحة الرئيسية)

Layout:

الأسبوع 17 - الثلث الثاني

بأقي 161 يوم حتى الموعد المتوقع

حجم طفلك: الرمانة

الطول: 13 سم

الوزن: 150 جرام

الوزن الحالي

الضغط الدم

كجم

68.5 120/80

طبيعي ✓

↑ +0.5

% التقدم في الثلث الثاني: 60

آخر تسجيل: منذ يومين

الملفات الطبية: 8

المراحل المكتملة: 3

[الأسابيع]

[إضافة تسجيل +]

Sections:

- 1. **Header:** Current week, trimester, countdown
- 2. **Baby Development Card:** Size, length, weight, cute illustration
- 3. **Quick Stats:** Weight, BP, symptoms
- 4. **Trimester Progress Bar**
- 5. **Recent Activity:** Last entry, files count, milestones
- 6. **Quick Actions:** Add entry, View weeks, Medical files

StartPregnancy.jsx (بدء حمل جديد)

Form:

- (LMP) آخر دورة شهرية - Date picker (required)

- Validation: not in future
- تاريخ الإخصاب (**Conception**) - Date picker (optional)
 - Auto-calculate: LMP + 14 days
- الموعد المتوقع للولادة - Date picker (optional)
 - Auto-calculate: LMP + 280 days
- ملاحظات - Textarea (optional)

Display:

- Explanation of LMP importance
- Visual timeline showing LMP → Conception → Due Date
- Submit button: "بدء متابعة الحمل"

On Success:

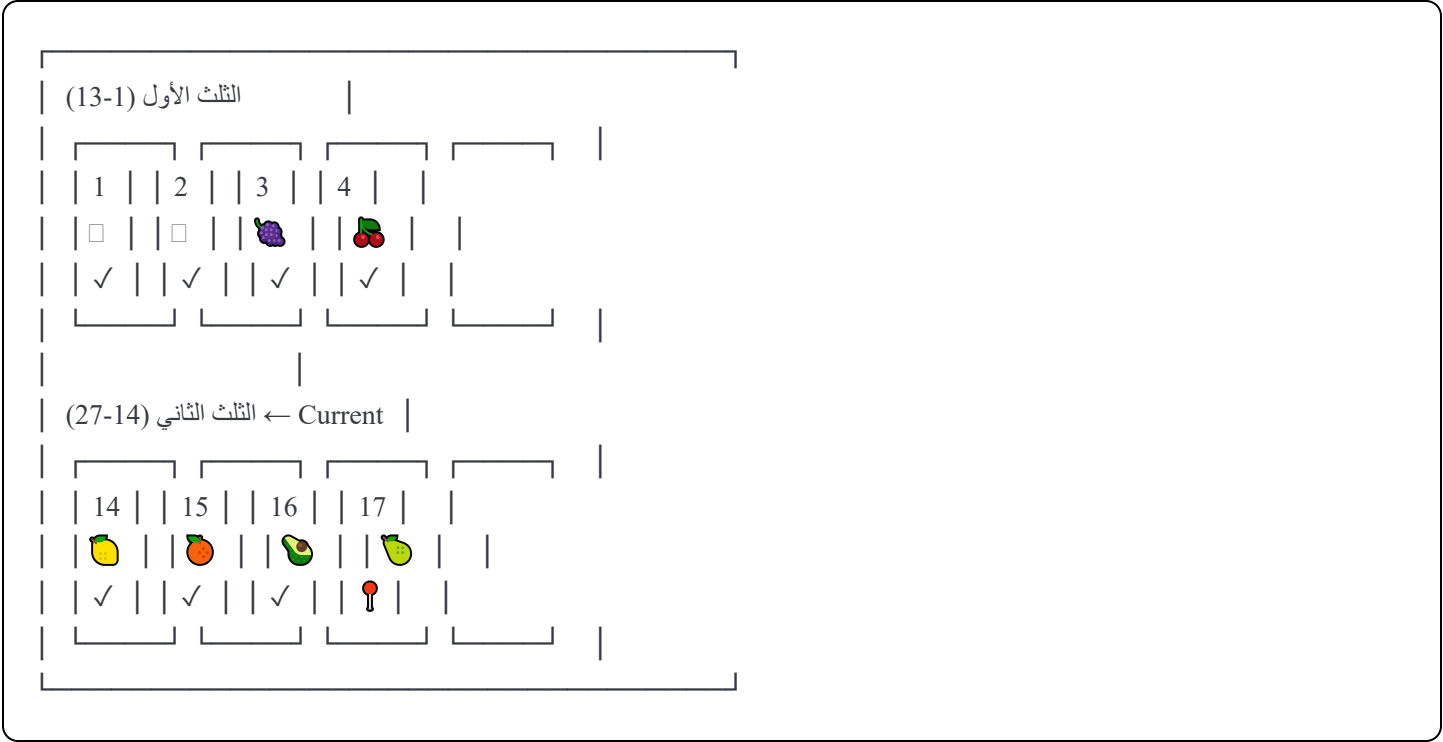
- Update life_stage to "pregnancy"
- Show congratulations message
- Navigate to PregnancyDashboard

WeekByWeek.jsx (التصفح حسب الأسبوع)

Features:

- **Horizontal Swiper** for 40 weeks
- Each card shows:
 - Week number
 - Baby size (fruit comparison)
 - Key development
 - Small image/illustration
- Highlight current week
- Mark completed weeks with ✓
- Tap card to see full details

Layout:



WeekDetails.jsx (تفاصيل أسبوع محدد)

Sections:

1. Header
 - Week number & trimester
 - Baby illustration
 - Size comparison
2. Baby Development (expandable)
 - Length & weight
 - Developmental milestones
 - What's forming this week
3. Mother's Changes (expandable)
 - Physical changes
 - Common symptoms
 - Body adaptations
4. Symptoms to Expect
 - List of common symptoms
 - How to manage each
5. Medical Tips

- What to eat
- What to avoid
- Medications (if any)

6. Nutrition & Exercise

- Recommended foods
- Safe exercises
- Foods to avoid

7. Warning Signs (always visible, red box)

- When to call doctor
- Emergency signs

8. Checklist

- Tests/appointments due
- Checkboxes to mark completion

9. Your Entries for This Week

- User's logged data
- Add new entry button

PregnancyLog.jsx (سجل التسجيلات)

Display:

- Timeline of all entries
- Group by week or month
- Each entry card shows:
 - Date & week number
 - Weight & BP
 - Symptoms
 - Notes
 - Edit/Delete buttons

Filters:

- By week range
- By trimester

- By symptom

Chart View:

- Weight progression
 - BP readings over time
-

AddEntry.jsx (إضافة/تعديل تسجيل)

Form Fields:

1. الأسبوع - Auto-filled, can change
2. الوزن - Number input (kg)
3. ضغط الدم - Two inputs (Systolic/Diastolic)
4. الأعراض - Multi-select with search
 - Predefined list: غثيان، صداع، آلام ظهر، حرقه معدة، دوخة، تورم، إلخ
 - Can add custom symptom
5. ملاحظات - Textarea
6. التاريخ - Date picker

Features:

- Real-time BP category (normal/elevated/high)
- Weight change from last entry
- Symptom suggestions based on week
- Save button with loading state

Alerts:

- Show warning if BP is high
 - Show concern if weight gain is too rapid/slow
-

WeightChart.jsx (رسم بياني للوزن)

Features:

- Line chart showing weight progression
- Starting weight line

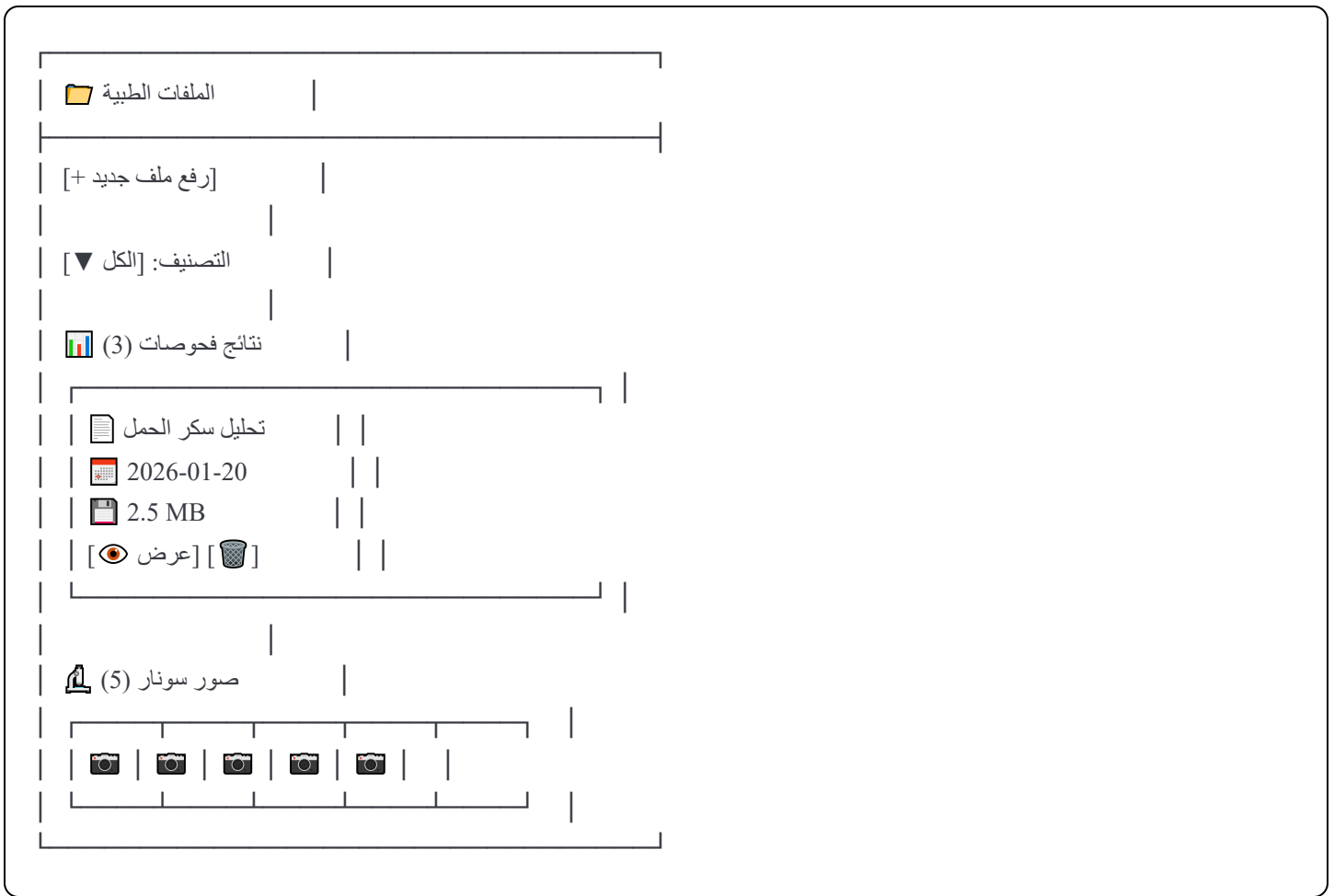
- Recommended weight range (shaded area)
- Current weight marker
- Weeks on X-axis, Weight on Y-axis
- Hover tooltips with exact values
- Color coding:
 - Green: within range
 - Orange: slightly above/below
 - Red: significantly out of range

Stats Display:

- Starting weight
 - Current weight
 - Total gain
 - Average weekly gain
 - Recommended total gain (based on BMI)
-

MedicalFiles.jsx (الملفات الطبية)

Layout:



Features:

- Upload with drag & drop
- Filter by category
- Gallery view for images
- List view for documents
- Preview modal for images/PDFs
- Download option
- Delete with confirmation
- Sort by date

Categories:

-  نتائج فحوصات (Lab Results)
-  صور سونار (Ultrasound)
-  روشتات (Prescriptions)
-  تقارير طبية (Medical Reports)
-  أخرى (Other)

PregnancyHistory.jsx (سجل الحمل السابقة)

Display:

- List of previous pregnancies
- Each card shows:
 - Pregnancy number
 - LMP & Due date
 - Delivery date & type
 - Duration (weeks)
 - Status badge
 - View details button

Stats:

- Total pregnancies
 - Completed pregnancies
 - Average duration
-

🧠 Key UI Components

PregnancyHeader.jsx

jsx

Props:

- `currentWeek` (number)
- `daysRemaining` (number)
- `trimester` (number)
- `babySize` (string)

Display:

- Large week number
- "يوم X باقي"
- Progress ring/circle
- Baby size illustration

BabyDevelopment.jsx

jsx

Props:

- `weekNumber` (number)
- `length` (number)
- `weight` (number)
- `sizeComparison` (string)
- `developments` (array)

Display:

- Cute illustration
- Size comparison
- Measurements
- Development list

SymptomTracker.jsx

jsx

Props:

- `selectedSymptoms` (array)
- `onSymptomsChange` (function)
- `commonSymptoms` (array)

Features:

- Grid of symptom chips
- Active/inactive states
- Custom symptom input
- Search functionality

TrimesterProgress.jsx

jsx

Props:

- `trimester` (number)
- `currentWeek` (number)
- `percentage` (number)

Display:

- 3 segments (trimester 1, 2, 3)
- Filled up to current progress
- Week markers
- Milestone indicators

MilestoneCard.jsx

jsx

Props:

- `week` (number)
- `title` (string)
- `description` (string)
- `completed` (boolean)
- `dueIn` (number)

Display:

- Icon/emoji
- Week number
- Title & description
- Status indicator
- Action button (if applicable)

API Service (pregnancyService.js)

javascript

```
export const pregnancyService = {  
  // Pregnancy management  
  startPregnancy: async (data) => {},  
  getCurrentPregnancy: async () => {},  
  completePregnancy: async (id, data) => {},  
  getPregnancyHistory: async () => {},  
  
  // Entries  
  addEntry: async (data) => {},  
  getEntries: async (params) => {},  
  updateEntry: async (id, data) => {},  
  deleteEntry: async (id) => {},  
  
  // Week information  
  getWeekInfo: async (weekNumber) => {},  
  getAllWeeksInfo: async () => {},  
  
  // Statistics  
  getStats: async () => {},  
  getWeightChart: async () => {},  
  
  // Medical files  
  uploadFile: async (file, metadata) => {},  
  getFiles: async (params) => {},  
  deleteFile: async (id) => {}  
};
```

☐ Custom Hook (usePregnancy.js)

```
javascript
```

```
export const usePregnancy = () => {
  const [pregnancy, setPregnancy] = useState(null);
  const [loading, setLoading] = useState(true);
  const [entries, setEntries] = useState([]);

  const startNewPregnancy = async (data) => {};
  const fetchCurrentPregnancy = async () => {};
  const addWeekEntry = async (data) => {};
  const completePregnancy = async (data) => {};
  const getWeekDetails = async (week) => {};

  // Auto-fetch on mount
  useEffect(() => {
    fetchCurrentPregnancy();
  }, []);

  return {
    pregnancy,
    entries,
    loading,
    startNewPregnancy,
    addWeekEntry,
    completePregnancy,
    getWeekDetails,
    // ... other functions
  };
};
```

Backend Business Logic

Date Calculations:

```
php
```

```

// Calculate current week from LMP
$Imp = Carbon::parse($pregnancy->last_menstrual_period);
$today = Carbon::today();
$daysPregnant = $Imp->diffInDays($today);
$currentWeek = (int) floor($daysPregnant / 7);

// Calculate due date (LMP + 280 days)
$dueDate = $Imp->copy()->addDays(280);

// Calculate trimester
if ($currentWeek <= 13) $trimester = 1;
elseif ($currentWeek <= 27) $trimester = 2;
else $trimester = 3;

// Days remaining
$daysRemaining = Carbon::today()->diffInDays($dueDate);

```

Weight Gain Recommendations:

```

php

// Based on pre-pregnancy BMI
if ($bmi < 18.5) {
    $recommendedGain = '12.5-18 kg';
} elseif ($bmi >= 18.5 && $bmi < 25) {
    $recommendedGain = '11-16 kg';
} elseif ($bmi >= 25 && $bmi < 30) {
    $recommendedGain = '7-11.5 kg';
} else {
    $recommendedGain = '5-9 kg';
}

```

Blood Pressure Categories:

```

php

```

```

$systolic = $entry->blood_pressure_systolic;
$diastolic = $entry->blood_pressure_diastolic;

if ($systolic < 120 && $diastolic < 80) {
    $category = 'normal';
} elseif ($systolic >= 120 && $systolic < 140 || $diastolic >= 80 && $diastolic < 90) {
    $category = 'elevated';
} else {
    $category = 'high';
}

// Alert if high
if ($category === 'high') {
    // Notify doctor / show warning
}

```

Baby Size Comparisons (per week):

```

php

$babySizes = [
    1 => ['comparison' => 'حجم بذرة', 'length' => 0, 'weight' => 0],
    4 => ['comparison' => 'حبة الخشخاش', 'length' => 0.2, 'weight' => 0],
    8 => ['comparison' => 'حبة الفاصوليا', 'length' => 1.6, 'weight' => 1],
    12 => ['comparison' => 'حبة الليمون', 'length' => 5.4, 'weight' => 14],
    16 => ['comparison' => 'حبة الأفوكادو', 'length' => 11.6, 'weight' => 100],
    20 => ['comparison' => 'حبة الموز', 'length' => 25.6, 'weight' => 300],
    24 => ['comparison' => 'كوز الذرة', 'length' => 30, 'weight' => 600],
    28 => ['comparison' => 'بادنجان', 'length' => 37.6, 'weight' => 1000],
    32 => ['comparison' => 'جوز الهند', 'length' => 42.4, 'weight' => 1700],
    36 => ['comparison' => 'رأس خس', 'length' => 47.4, 'weight' => 2600],
    40 => ['comparison' => 'بطيخة صغيرة', 'length' => 51.2, 'weight' => 3400],
];

```

📅 Week Content Database

تحتوي على JSON يفضل إنشاء جدول أو ملفات

- معلومات تطور الجنين لكل أسبوع
- التغيرات في جسم الأم
- الأعراض الشائعة

- النصائح الطبية
- النصائح الغذائية
- التمارين الآمنة

Option 1: Database table `pregnancy_weeks_content` **Option 2:** JSON files `/storage/pregnancy-content/week-{number}.json` **Option 3:** Hardcoded in backend (not recommended)

Important Features

Alerts & Notifications:

1. **High Blood Pressure** → Notify doctor immediately
2. **Rapid Weight Gain** → Show warning, suggest doctor visit
3. **Warning Symptoms** → List of red flags (bleeding, severe pain, etc.)
4. **Milestone Reminders** → Upcoming tests/appointments

Security:

- Patient can only view/edit own pregnancy data
- Soft delete for sensitive data
- File uploads: strict validation
- HIPAA-compliant file storage

Performance:

- Lazy load week content
 - Cache week information
 - Optimize image uploads (compress, resize)
 - Pagination for entries
-

Testing Checklist

Backend:

- ☐ Start pregnancy correctly calculates due date
- ☐ Current week calculation accurate
- ☐ Weight/BP entries save correctly

- ☐ File uploads work and validate
- ☐ Statistics calculations accurate
- ☐ Only patient can access own data
- ☐ Soft delete works

Frontend:

- ☐ Dashboard shows correct info
 - ☐ Week navigation smooth
 - ☐ Charts render correctly
 - ☐ Forms validate properly
 - ☐ File upload/delete works
 - ☐ Responsive on mobile
 - ☐ Loading states display
 - ☐ Error handling works
-



Implementation Priority

Phase 1: Core Pregnancy Tracking

1. Start pregnancy
2. Get current pregnancy
3. Add/view entries
4. Basic dashboard

Phase 2: Week Information

1. Week details page
2. Week-by-week navigation
3. Content for all 40 weeks

Phase 3: Advanced Features

1. Weight chart
2. Medical files
3. Statistics
4. Pregnancy history

Phase 4: Polish

1. Animations
 2. Better UI/UX
 3. Performance optimization
 4. Comprehensive testing
-

Important Notes

1. **Sensitivity:** Pregnancy data is extremely sensitive - handle with care
 2. **Medical Accuracy:** All medical information must be reviewed by medical professionals
 3. **Disclaimers:** Always include disclaimer that app is not a substitute for medical advice
 4. **Emergency Info:** Prominently display warning signs and when to seek immediate care
 5. **Positive Messaging:** Use encouraging, supportive language throughout
 6. **Cultural Sensitivity:** Content appropriate for Egyptian/Arabic culture
-

Success Criteria

System is complete when:

-  User can start and track pregnancy
 -  Weekly entries work smoothly
 -  All 40 weeks have complete information
 -  Charts display accurately
 -  Medical files upload/download
 -  Mobile-friendly interface
 -  Fast and responsive
 -  Secure and private
-

Note: هذا نظام كبير ومهم جدًا. يجب مراجعة المحتوى الطبي مع متخصصين قبل النشر.